

Networking Basics Guide

Beginner-Friendly Practical Networking for Cybersecurity, IT & Networking

Dark-themed practical handbook with diagrams, networking tools, attack surfaces, subnetting basics, and cybersecurity-focused explanations.

Table of Contents

1. Introduction to Networking
2. Ports & Services
3. Protocols
4. TCP vs UDP
5. Subnetting Basics
6. Common Attack Surfaces
7. Practical Networking Tools
8. Final Summary

1. Introduction to Networking

Networking is the process of devices communicating with each other to exchange data. Devices such as computers, phones, servers, routers, and IoT devices communicate through wired or wireless connections.

A network allows systems to share internet access, files, printers, databases, and services. In cybersecurity, understanding networking is essential because attacks and defenses both operate across networks.

Simple Device Communication



LAN (Local Area Network) connects devices in a small area such as a home, office, or school.

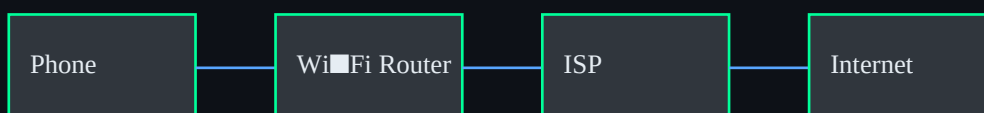
WAN (Wide Area Network) connects multiple LANs across large distances. The internet is the largest WAN.

Quick Tip: Most home networks use private IP addresses internally and one public IP address externally.

IP Address: A logical address used to identify devices on a network. Example: 192.168.1.5

MAC Address: A physical hardware address assigned to a network card. Example: 00:1A:2B:3C:4D:5E

Home Network Example



Office Network Example



2. Ports & Services

Ports are logical communication endpoints used by applications and services. A device can have thousands of ports, each linked to a specific service.

Attackers frequently scan systems for open ports because exposed services may contain vulnerabilities or weak configurations.

Port	Service	Protocol	Purpose
20/21	FTP	TCP	File Transfer
22	SSH	TCP	Secure Remote Login
23	Telnet	TCP	Insecure Remote Login
25	SMTP	TCP	Email Transfer
53	DNS	TCP/UDP	Name Resolution
80	HTTP	TCP	Web Traffic
443	HTTPS	TCP	Encrypted Web Traffic
445	SMB	TCP	Windows File Sharing
3306	MySQL	TCP	Database Access
3389	RDP	TCP	Remote Desktop

Quick Tip: Use Nmap to identify exposed ports during security assessments.

Example Nmap Scan:

```
nmap -sV 192.168.1.10
```

This scans for open ports and service versions.

Cybersecurity Example: Attackers often target RDP (3389) for brute-force attacks and SMB (445) for ransomware propagation.

3. Protocols

HTTP/HTTPS

Purpose: Web communication

Security Risk: Unencrypted HTTP traffic can be intercepted

DNS

Purpose: Domain lookup

Security Risk: DNS spoofing and cache poisoning

DHCP

Purpose: Automatic IP assignment

Security Risk: Rogue DHCP servers

FTP

Purpose: File transfers

Security Risk: Plain-text credentials

SSH

Purpose: Secure remote administration

Security Risk: Weak passwords and outdated versions

SMTP

Purpose: Email delivery

Security Risk: Spam and phishing

TCP

Purpose: Reliable communication

Security Risk: Session hijacking

UDP

Purpose: Fast lightweight communication

Security Risk: Amplification attacks

ICMP

Purpose: Diagnostics and ping

Security Risk: Reconnaissance

ARP

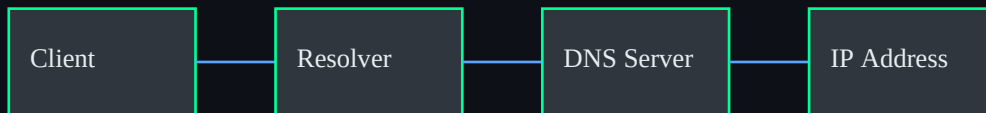
Purpose: Local network mapping

Security Risk: ARP spoofing

HTTP Request Flow



DNS Resolution Flow



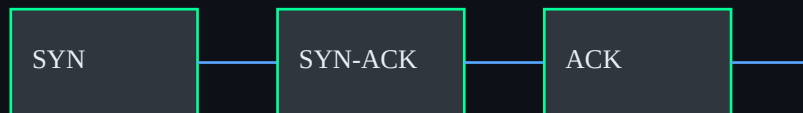
Quick Tip: HTTPS encrypts traffic using TLS, protecting credentials and sensitive data.

4. TCP vs UDP

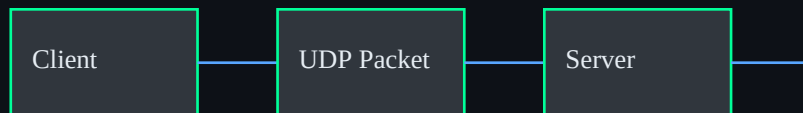
Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable	Less reliable
Speed	Slower	Faster
Error Checking	Strong	Minimal
Common Uses	Web, Email	Gaming, Streaming

TCP is like a phone call where both sides confirm communication. UDP is like broadcasting a message without waiting for confirmation.

TCP 3-Way Handshake



UDP Communication



Quick Tip: Streaming and gaming often prefer UDP because speed matters more than perfect delivery.

5. Subnetting Basics

Subnetting divides networks into smaller logical sections. This improves organization, performance, and security.

CIDR Examples:

/24 = 255.255.255.0

/16 = 255.255.0.0

/8 = 255.0.0.0

Private IP Ranges:

10.0.0.0/8

172.16.0.0 – 172.31.255.255

192.168.0.0/16

Subnet Split Example



Quick Tip: A /24 subnet usually supports 254 usable hosts.

Easy Trick: Larger CIDR number = smaller network size.

Exercise	Answer
How many usable hosts in /24?	254
What mask is /16?	255.255.0.0
Is 192.168.1.5 public or private?	Private

6. Common Attack Surfaces

Open Ports

How Attackers Exploit It: Attackers scan for vulnerable services

Basic Defense: Close unused ports

Weak Passwords

How Attackers Exploit It: Brute-force attacks

Basic Defense: Use MFA and strong passwords

Public Wi-Fi

How Attackers Exploit It: Traffic interception

Basic Defense: Use VPNs

Outdated Services

How Attackers Exploit It: Known exploits

Basic Defense: Patch regularly

DNS Spoofing

How Attackers Exploit It: Redirect users to fake sites

Basic Defense: Use DNS security tools

MITM Attacks

How Attackers Exploit It: Intercept communication

Basic Defense: Use HTTPS and encryption

MITM Attack Flow



Quick Tip: Disable services you do not use. Every open service increases attack surface.

Security Checklist

Change default router passwords

Enable firewall

Update firmware

Use WPA2/WPA3

Disable unused services

7. Practical Networking Tools

Tool	Purpose	Example
ping	Tests connectivity	ping google.com
tracert	Shows network path	tracert google.com
ipconfig/ifconfig	Displays network config	ipconfig
netstat	Shows active connections	netstat -an
nslookup	Queries DNS	nslookup openai.com
dig	Advanced DNS lookup	dig openai.com
nmap	Scans ports/services	nmap -sV 192.168.1.1
Wireshark	Captures packets	Analyze traffic visually

Troubleshooting Scenario: A website does not load. Use ping to test connectivity, nslookup to verify DNS, and tracert to identify routing problems.

Quick Tip: Wireshark is one of the most valuable tools for learning how real traffic behaves on networks.

8. Final Summary

Networking forms the foundation of cybersecurity. Understanding IP addresses, protocols, ports, routing, and traffic analysis allows you to understand how attacks work and how defenses are implemented.

Cybersecurity Mindset:

- Assume every exposed service can become a target
- Visibility matters: monitor traffic and logs
- Least privilege reduces damage
- Segmentation improves containment

Recommended Beginner Labs:

- Build a home lab using VirtualBox or VMware
- Scan your own network using Nmap
- Capture packets using Wireshark
- Configure a firewall rule
- Practice subnetting exercises daily

Quick Tip: The fastest way to learn networking is to build and troubleshoot your own lab environment.

Condensed Quick Reference

Concept	Quick Recall
IP Address	Logical network identity
MAC Address	Physical hardware identity
TCP	Reliable communication
UDP	Fast communication
DNS	Translates domains to IPs
SSH	Secure remote access
HTTPS	Encrypted web traffic